

MAT 535 HW06

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Problem 1 (§14.2#5). Prove that the Galois group of $x^p - 2$ is isomorphic to the group of matrices

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a, b \in \mathbb{F}_p, a \neq 0 \right\}.$$

Solution. Observe that the splitting field of $x^p - 2$ must contain $\sqrt[p]{2}$, whose minimal polynomial has degree p , and a primitive p th root of unity ζ , whose minimal polynomial has degree $p - 1$. Since $x^p - 2$ splits in $\mathbb{Q}(\sqrt[p]{2}, \zeta)$, this extension must be a splitting field for $x^p - 2$, and $[\mathbb{Q}(\sqrt[p]{2}, \zeta) : \mathbb{Q}] = p(p - 1)$ because the degree is bounded above by $p(p - 1)$ and divisible by p and $p - 1$.

The Galois group of this splitting extension is generated by the $\mathbb{Q}$ -automorphisms ρ and σ defined by

$$\begin{aligned} \rho(\sqrt[p]{2}) &= \zeta \sqrt[p]{2}, & \rho(\zeta) &= \zeta, \\ \sigma(\sqrt[p]{2}) &= \sqrt[p]{2}, & \sigma(\zeta) &= \zeta^g \end{aligned}$$

where g is chosen so that $\zeta \mapsto \zeta^g$ generates the cyclic group $\text{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$, or equivalently g generates $(\mathbb{Z}/p\mathbb{Z})^\times$. The defining relations between σ and ρ are

$$\rho^p = \sigma^{p-1} = 1, \quad \sigma \rho \sigma^{-1} = \rho^g$$

(these determine the Galois group uniquely because the Sylow theorems imply that the subgroup of order p is normal, so that the Galois group is a semidirect product $\mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/(p - 1)\mathbb{Z}$).

Consider the map $\Phi: \text{Gal}(\mathbb{Q}(\sqrt[p]{2}, \zeta)/\mathbb{Q}) \rightarrow G$ sending

$$\Phi(\rho) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \Phi(\sigma) = \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}.$$

To check that this is an isomorphism we just need to check that Φ commutes with the defining relations, since this implies that G has the same presentation as the Galois group.

$$\Phi(\rho^k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

for any integer k implies that $\Phi(\rho)$ has order p ;

$$\Phi(\sigma^k) = \begin{pmatrix} g^k & 0 \\ 0 & 1 \end{pmatrix}$$

implies that $\Phi(\sigma)$ has order $p - 1$; and

$$\Phi(\sigma \rho \sigma^{-1}) = \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & g \\ 0 & 1 \end{pmatrix} = \Phi(\rho^g).$$

□

Problem 2 (§14.2#10). Determine the Galois group over $\mathbb{Q}$ of $x^8 - 3$.

Solution. The splitting field of $x^8 - 3$ is the degree 32 extension $E = \mathbb{Q}(\sqrt[8]{3}, \zeta)$ where $\zeta = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}$ is a primitive 8th root of unity; the extension has degree 32 because $[E : \mathbb{Q}(\sqrt[8]{3})] = 4$, since the smaller field only contains real numbers.

Since $E/\mathbb{Q}(\sqrt[8]{3})$ is a biquadratic extension generated by i and $\sqrt{2}$,

$$\text{Gal}(E/\mathbb{Q}(\sqrt[8]{3})) \cong V_4 = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z},$$

consisting of the automorphisms σ given by complex conjugation (hence swapping $\zeta \leftrightarrow \zeta^7$ and $\zeta^3 \leftrightarrow \zeta^5$); τ given by negating ζ (hence swapping $\zeta \leftrightarrow \zeta^5$ and $\zeta^3 \leftrightarrow \zeta^7$); and $\sigma\tau$ (which swaps $\zeta \leftrightarrow \zeta^3$ and $\zeta^5 \leftrightarrow \zeta^7$).

We also see that $\text{Gal}(E/\mathbb{Q}(\zeta))$ contains the automorphism φ sending $\sqrt[8]{3} \mapsto \zeta \cdot \sqrt[8]{3}$. The powers of φ send $\varphi^k(\sqrt[8]{3}) = \zeta^k \sqrt[8]{3}$, so φ has order 8, meaning it generates the order 8 group $\text{Gal}(E/\mathbb{Q}(\zeta)) \cong \mathbb{Z}/8\mathbb{Z}$. Additionally, $\mathbb{Q}(\zeta)/\mathbb{Q}$ is a Galois extension (since $\mathbb{Q}(\zeta)$ is the splitting field of $x^4 + 1$), so $\text{Gal}(E/\mathbb{Q}(\zeta))$ is normal in $\text{Gal}(E/\mathbb{Q})$.

Finally, observe that $\text{Gal}(E/\mathbb{Q}(\zeta)) \cap \text{Gal}(E/\mathbb{Q}(\sqrt[8]{3}))$ fixes the compositum $\mathbb{Q}(\zeta)\mathbb{Q}(\sqrt[8]{3}) = E$, so the intersection of these subgroups is the trivial subgroup, while $\text{Gal}(E/\mathbb{Q}(\zeta)) \cdot \text{Gal}(E/\mathbb{Q}(\sqrt[8]{3}))$ fixes $\mathbb{Q}(\zeta) \cap \mathbb{Q}(\sqrt[8]{3}) = \mathbb{Q}$, hence equals $\text{Gal}(E/\mathbb{Q})$. This means $\text{Gal}(E/\mathbb{Q}) \cong \mathbb{Z}/8\mathbb{Z} \rtimes_{\theta} V_4$, with the action θ of V_4 on $\mathbb{Z}/8\mathbb{Z}$ given by

$$\begin{aligned} \theta_{\sigma}(\varphi) &= \sigma\varphi\sigma = \begin{cases} \zeta & \mapsto \zeta \\ \sqrt[8]{3} & \mapsto \zeta^7 \cdot \sqrt[8]{3} \end{cases} \\ \theta_{\tau}(\varphi) &= \tau\varphi\tau = \begin{cases} \zeta & \mapsto \zeta \\ \sqrt[8]{3} & \mapsto \zeta^5 \cdot \sqrt[8]{3} \end{cases} \\ \theta_{\sigma\tau}(\varphi) &= (\sigma\tau)\varphi(\sigma\tau) = \begin{cases} \zeta & \mapsto \zeta \\ \sqrt[8]{3} & \mapsto \zeta^3 \cdot \sqrt[8]{3}, \end{cases} \end{aligned}$$

or equivalently,

$$\text{Gal}(E/\mathbb{Q}) = \langle \sigma, \tau, \varphi \mid \varphi^8 = \sigma^2 = \tau^2 = 1, \sigma\varphi\sigma = \varphi^7, \tau\varphi\tau = \varphi^5 \rangle$$

(noting that this implies $(\sigma\tau)\varphi(\sigma\tau) = \tau(\sigma\varphi\sigma)\tau = \tau\varphi^7\tau = \varphi^{35} = \varphi^3$). □

Problem 3 (§14.2#16). (a) Prove that $x^4 - 2x^2 - 2$ is irreducible over $\mathbb{Q}$.

(b) Show that the roots of this quartic are

$$\alpha_1 = \sqrt{1 + \sqrt{3}}, \quad \alpha_2 = \sqrt{1 - \sqrt{3}}, \quad \alpha_3 = -\sqrt{1 + \sqrt{3}}, \quad \alpha_4 = -\sqrt{1 - \sqrt{3}}.$$

(c) Let $K_1 = \mathbb{Q}(\alpha_1)$ and $K_2 = \mathbb{Q}(\alpha_2)$. Show that $K_1 \neq K_2$ and $K_1 \cap K_2 = \mathbb{Q}(\sqrt{3}) = F$.

(d) Show that K_1 , K_2 , and K_1K_2 are Galois over F with $\text{Gal}(K_1K_2/F)$ the Klein 4-group. Write out the elements of $\text{Gal}(K_1K_2/F)$ explicitly. Determine all the subgroups of the Galois group and give their corresponding fixed subfields of K_1K_2 containing F .

(e) Prove that the splitting field of $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ is of degree 8 with dihedral Galois group.

Solution. (a) Follows from Eisenstein's criterion with the prime 2.

(b) Follows from writing $x^4 - 2x^2 - 2 = (x^2 - 1)^2 - 3$, since setting this equal to zero implies $x^2 - 1 = \pm\sqrt{3}$.

(c) On one hand, $K_1 \cap K_2$ contains $\alpha_1^2 - 1 = 1 - \alpha_2^2 = \sqrt{3}$, so it contains $\mathbb{Q}(\sqrt{3})$. On the other hand, $K_1 \cap K_2$ is a proper subextension of $K_1/\mathbb{Q}$, so its degree over $\mathbb{Q}$ is a proper divisor of $[K_1 : \mathbb{Q}] = 4$, meaning the degree is at most 2. This shows that $K_1 \cap K_2$ cannot be larger than $\mathbb{Q}(\sqrt{3})$.

(d) We can see that $K_1K_2 = \mathbb{Q}(\alpha_1, \alpha_2)$ is a splitting field of $x^4 - 2x^2 - 2$, so certainly K_1K_2/F is a Galois extension. Since

$$2 = [K_1 : F] < [K_1K_2 : F] \leq [K_1 : F][K_2 : F] = 4$$

and $[K_1K_2 : F]$ is divisible by $[K_1 : F] = 2$, the degree of K_1K_2/F is 4. Furthermore $\text{Gal}(K_1K_2/F)$ contains the two order-2 automorphisms σ and τ defined by

$$\begin{aligned} \sigma(\alpha_1) &= \alpha_3, & \sigma(\alpha_2) &= \alpha_2, & \sigma(\alpha_3) &= \alpha_1, & \sigma(\alpha_4) &= \alpha_4, \\ \tau(\alpha_1) &= \alpha_1, & \tau(\alpha_2) &= \alpha_4, & \tau(\alpha_3) &= \alpha_3, & \tau(\alpha_4) &= \alpha_2. \end{aligned}$$

To check that σ is an automorphism, we note that σ fixes K_2 and swaps the roots of the polynomial $x^2 - 1 - \sqrt{3}$, hence factors through the map $K_2[x] \rightarrow K_2(\alpha_1) = K_1K_2$ sending x to α_1 , so that σ is a field homomorphism (hence isomorphism, by degree considerations) $K_2[x]/(x^2 - 1 - \sqrt{3}) \cong K_1K_2 \rightarrow K_1K_2$; similar reasoning shows that τ is a field isomorphism.

The only groups of order 4 up to isomorphism are $\mathbb{Z}/4\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, but the former only has one element of order 2, so $\text{Gal}(K_1K_2/F) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (and the remaining non-identity automorphism is $\sigma\tau = \tau\sigma$, which swaps both pairs $\{\alpha_1, \alpha_3\}$ and $\{\alpha_2, \alpha_4\}$).

Since $G = \text{Gal}(K_1K_2/F)$ is abelian, its subgroups are all normal subgroups of G , meaning the subextensions K_1/F and K_2/F of K_1K_2/F are both Galois extensions, by the fundamental theorem. Furthermore, the subgroups of G are $\{1\}$, $\{1, \sigma\}$, $\{1, \tau\}$, $\{1, \sigma\tau\}$, and G , which respectively have fixed fields K_1K_2 , K_2 , K_1 , $F(\alpha_1\alpha_2) = F(\sqrt{-2})$, and F .

(e) The degree of the splitting field $E = K_1K_2$ of $x^4 - 2x^2 - 2$ over $\mathbb{Q}$ is $[E : F][F : \mathbb{Q}] = 4 \cdot 2 = 8$. Consider $\text{Gal}(E/\mathbb{Q})$ as a subgroup of S_4 , with $k = 1, 2, 3, 4$ corresponding to α_k (so σ from part (d) is the permutation $(1\ 3)$). Then the permutations in (d) are

$$1 = (1), \quad \sigma = (1\ 3), \quad \tau = (2\ 4), \quad \sigma\tau = \tau\sigma = (1\ 3)(2\ 4).$$

The remaining four automorphisms are the ones not fixing F ; one such automorphism is $(1\ 2)(3\ 4)$. This is an automorphism because it fixes the field $\mathbb{Q}(\beta)$ where $\beta^2 = 2 + 2\sqrt{-2}$ and swaps the roots of $x^2 - \beta x + \sqrt{-2} = (x - \alpha_1)(x - \alpha_2)$ (similar to the logic for σ being an automorphism in (d)). The coset of $\text{Gal}(E/F)$ by (the automorphism corresponding to) $(1\ 2)(3\ 4)$ is $\{(1\ 2)(3\ 4), (1\ 2\ 3\ 4), (1\ 4\ 3\ 2), (1\ 4)(2\ 3)\}$. Thus, $\text{Gal}(E/\mathbb{Q})$ is generated as a subgroup of S_4 by $\varphi = (1\ 2\ 3\ 4)$ and σ , which satisfy $\varphi^4 = \sigma^2 = (\sigma\varphi)^2 = 1$, meaning it is dihedral. $\square$

Problem 4 (14.4#5). Let p be a prime and let F be a field. Let K be a Galois extension of F whose Galois group is a p -group. Such an extension is called a p -extension.

- (a) Let L be a p -extension of K . Prove that the Galois closure of L over F is a p -extension of F .
- (b) Give an example to show that (a) need not hold if $[K : F]$ is a power of p but K/F is not Galois.

Solution. (a) Let $\bar{F}$ be the Galois closure of L over F and define $G = \text{Gal}(\bar{F}/F)$, $N = \text{Gal}(\bar{F}/K)$, and $Q = \text{Gal}(\bar{F}/L)$. The given setup translates to the following group-theoretic information:

- $N \trianglelefteq G$ and $Q \trianglelefteq N$, since K/F and L/K are Galois;
- $(G : N) = [K : F]$ and $(N : Q) = [L : K]$ are both powers of p ; and
- Q does not contain any nontrivial normal subgroups of G , since no proper L -subextension of $\bar{F}$ is Galois over F .

Our goal is to show that G (and Q and N) are p -groups.

Let $Q = Q_1, Q_2, \dots, Q_n$ be the distinct conjugate subgroups of Q in G , so

$$\{Q_1, \dots, Q_n\} = \{gQg^{-1} : g \in G\}.$$

Since N is normal in G and Q is normal in N , the conjugates of Q are each normal subgroups of N . Furthermore, since Q does not contain any nontrivial subgroups of G and the intersection of the Q_i is by definition normal in G , we find that $Q_1 \cap \dots \cap Q_n = \{1\}$. Then, recalling that in general $|H \cdot K| = \frac{|H||K|}{|H \cap K|}$ for subgroups H, K of a finite group, we can compute

$$(G : Q_1 \cap \dots \cap Q_k) = \frac{(G : Q_1 \cap \dots \cap Q_{k-1})(G : Q_k)}{(G : (Q_1 \cap \dots \cap Q_{k-1}) \cdot Q_k)},$$

so in particular $(G : Q_1 \cap \dots \cap Q_k)$ divides $(G : Q_1 \cap \dots \cap Q_{k-1})(G : Q_k)$. Since $(G : Q_i)$ is a power of p for each $i = 1, \dots, n$, we have shown by induction that $(G : Q_1 \cap \dots \cap Q_k)$ is a (factor of a) power of p for each $k = 1, \dots, n$, so $(G : Q_1 \cap \dots \cap Q_n) = (G : 1) = |G|$ is a power of p .

(b) For odd primes p , let $F = \mathbb{Q}$ and $K = L = \mathbb{Q}(\sqrt[p]{2})$. We can see that $[K : F] = p$ and K/F is not a Galois extension (since $x^p - 2$ has a root in K but doesn't split), and L/K is trivially a p -extension. Then the Galois closure of L over $\mathbb{Q}$ is a subextension of $E = L(\zeta)$ which strictly contains L , where ζ is a root of $1 + x + \dots + x^{p-1}$, since $\mathbb{Q}(\sqrt[p]{2}, \zeta)$, being the splitting field of $x^p - 2$, is Galois over $\mathbb{Q}$. Since the degree of $\bar{F}$ in L divides $[E : L] = p - 1$ and is not equal to 1, $\bar{F}$ cannot be a p -extension. $\square$

Problem 5 (§14.4#8). Let K_1 and K_2 be two algebraic extensions of a field F contained in the field L of characteristic zero. Prove that the F -algebra $K_1 \otimes_F K_2$ has no nonzero nilpotent elements.

Solution. Consider an arbitrary element $\sum_{i=1}^n \alpha_i \otimes \beta_i \in K_1 \otimes_F K_2$ for $\alpha_i \in K_1$ and $\beta_i \in K_2$. Then in fact c lies in the sub- F -algebra $K_1 \otimes_F F(\beta_1, \dots, \beta_n)$. Since L has characteristic zero, $F(\beta_1, \dots, \beta_n)$ is a separable, finite extension, so it is a simple extension $F(\theta)$ by the primitive element theorem.

Lemma 5.1 (§14.4#7). Let $F \subseteq K \subseteq L$ and let $\theta \in L$ with $p(x) = m_{\theta, F}(x)$. Prove that $K \otimes_F F(\theta) \cong K[x]/(p(x))$ as K -algebras.

Proof. Let $d = \deg p$, so $F(\theta)$ has F -basis $1, \theta, \dots, \theta^{d-1}$. Then consider the map $\varphi: K[x] \rightarrow K \otimes_F F(\theta)$ defined by

$$\varphi(a_0 + a_1x + \dots + a_mx^m) = a_0 \otimes 1 + a_1 \otimes \theta + \dots + a_m \otimes \theta^m.$$

To verify that this is a K -algebra homomorphism we note that $\varphi(1) = 1 \otimes 1$ is the identity,

$$\varphi(ax^p \cdot bx^q) = ab \otimes x^{p+q} = (a \otimes x^p) \cdot (b \otimes x^q)$$

so that φ commutes with multiplication of monomials, and then it commutes with multiplication of arbitrary polynomials using K -linearity. Additionally, φ is clearly surjective.

The kernel of φ contains $p(x)$ because

$$\varphi(p(x)) = 1 \otimes p(\theta) = 0,$$

noting that $p(x) \in F[x]$ means that we can move the coefficients of p through the $\otimes$. On the other hand, since $F(\theta)$ is isomorphic to F^d as an F -vector space, $K \otimes_F F(\theta) \cong K^d$ as a K -vector space, which means that $K[x]/\ker \varphi$ must be a d -dimensional K -vector space. This forces the generator of $\ker \varphi$ to be a polynomial of degree d , so $\ker \varphi = (p(x))$, implying $K[x]/(p(x)) \cong K \otimes_F F(\theta)$ as desired. $\square$

The lemma implies that there is a K -algebra isomorphism between $K_1 \otimes_F F(\theta)$ and $K[x]/(p(x))$. The latter has no nonzero nilpotent elements (because it is a field), so the former also has no nonzero nilpotent elements; thus the arbitrarily chosen element $\sum \alpha_i \otimes \beta_i$ cannot be nilpotent unless it is zero. $\square$